

MTH2021 Mathematical Methods 2

Self-study sheet 1 – solutions

1.

$$z_1 + 2z_2 = 2 + 3i + 2(1 - 2i) = 2 + 3i + 2 - 4i = 4 - i \quad (1)$$

$$z_1 z_2 = (2 + 3i)(1 - 2i) = 2 + 3i - 4i - 6i^2 = 8 - i \quad (2)$$

$$\frac{z_1}{z_2} = \frac{z_1 z_2^*}{z_2 z_2^*} = \frac{(2 + 3i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{-4 + 7i}{5} \quad (3)$$

Let's evaluate the modulus of z_1/z_2 in two ways. First,

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} = \frac{\sqrt{2^2 + 3^2}}{\sqrt{1^2 + (-2)^2}} = \sqrt{13/5} \quad (4)$$

We get the same answer from $|z_1/z_2| = \sqrt{(-4/5)^2 + (7/5)^2}$. For the principal value of the argument of z_1/z_2 , since $\text{Re}(z_1/z_2) < 0$ and $\text{Im}(z_1/z_2) > 0$, we have

$$\text{Arg}(z_1/z_2) = \text{Arctan}(-7/4) + \pi = 2.0899 \quad (5)$$

(Arctan denotes the principal value of the inverse tangent: an angle in the range $(-\pi/2, \pi/2)$.)

2.

(a) $\arg z = 1237.2\pi = 618 \times 2\pi + 1.2\pi$ so we can choose the simpler $\arg z = 1.2\pi$. But this is still not in $(-\pi, \pi]$. (The square bracket on the right means that π is included in the interval.) To bring it into that range, we choose $\arg z = 1.2\pi - 2\pi = -0.8\pi$: $\text{Arg} z = -0.8\pi$.

(b)

$$z = \sqrt{3} + i = 2(\sqrt{3}/2 + i/2) = 2(\cos(\pi/6) + i \sin(\pi/6)) \quad (6)$$

$$\Rightarrow z^8 = 2^8 (\cos(8\pi/6) + i \sin(8\pi/6)) = 256 (\cos(4\pi/3) + i \sin(4\pi/3)) \quad (7)$$

$$\Rightarrow |z^8| = 256, \quad \text{Arg}(z^8) = -2\pi/3 \quad (8)$$

For the multi-valued $\arg(z^8)$ we can choose $\text{Arg}(z^8) + 2k\pi$ for any integer k . E.g. for $k = 1$ we get back $\arg(z^8) = -2\pi/3 + 2\pi = 4\pi/3$.

3.

(a) We have $-i = \exp(-i\pi/2)$ and

$$z^3 = 3^3 \exp(i(-\pi/2 + 2k\pi)) \Rightarrow z = 3 \exp(i(-\pi/2 + 2k\pi)/3) \quad (9)$$

Distinct solutions:

$$k = 0: z = 3 \exp(-i\pi/6) = 3(\cos(\pi/6) - i \sin(\pi/6)) = 3\sqrt{3}/2 - 3i/2 \quad (10)$$

$$k = 1: z = 3 \exp(i\pi/2) = 3i \quad (11)$$

$$k = 2: z = 3 \exp(7\pi i/6) = -3\sqrt{3}/2 - 3i/2 \quad (12)$$

For the second example:

$$z^3 = \exp(2k\pi i) \Rightarrow z = \exp(2k\pi i/3) \quad (13)$$

Distinct solutions:

$$k = 0 : z = 1 \quad (14)$$

$$k = 1 : z = \exp(2\pi i/3) = -1/2 + i\sqrt{3}/2 \quad (15)$$

$$k = 2 : z = \exp(4\pi i/3) = -1/2 - i\sqrt{3}/2 \quad (16)$$

For the third example:

$$z^4 = \exp(i(\pi/2 + 2k\pi)) \Rightarrow z = \exp(i(\pi/2 + 2k\pi)/4) \quad (17)$$

with distinct solutions for $k = 0, 1, 2, 3$.

(b) The equation

$$z(1 + z + z^2 + z^3 + z^4) = 0 \quad (18)$$

has the obvious solution $z = 0$. (**NB:** to cancel a factor from both sides of an equation, this factor must be non-zero. Thus, before cancelling anything, we must consider the case when it's zero.) Having counted it, we can cancel the z , leaving

$$1 + z + z^2 + z^3 + z^4 = 0 \quad (19)$$

Using the formula for a geometric series:

$$1 + z + z^2 + z^3 + z^4 = \frac{1 - z^5}{1 - z} = 0 \quad (20)$$

To keep the fraction defined, we assume $z \neq 1$. (We are not losing solutions for z this way, because $z = 1$ is not a solution to the starting equation.) We can now multiply both sides by $1 - z$:

$$1 - z^5 = 0 \Rightarrow z^5 = 1 \quad (21)$$

and we have reduced the problem to finding the fifth roots of 1:

$$z = \exp(2k\pi i/5) = 1, \exp(2\pi i/5), \exp(4\pi i/5), \exp(6\pi i/5), \exp(8\pi i/5) \quad (22)$$

But $z = 1$ we've excluded. So the solutions (including the original one above) are:

$$z = 0, \exp(2\pi i/5), \exp(4\pi i/5), \exp(6\pi i/5), \exp(8\pi i/5) \quad (23)$$

4.

(a) $z = t + it^3 = x + iy \Rightarrow y = t^3 = x^3$; this is the cubic $y = x^3$ (over all x).

(b) Putting $z = x + iy$:

$$|z - i|^2 + |z + i|^2 = 4 \quad (24)$$

$$(x^2 + (y - 1)^2) + (x^2 + (y + 1)^2) = 4 \quad (25)$$

$$2x^2 + 2y^2 + 2 = 4 \Rightarrow x^2 + y^2 = 1 \quad (26)$$

This is the unit circle centered at the origin.

(c) $z - i = x + i(y - 1)$ and thus

$$|z - i| = \sqrt{x^2 + (y - 1)^2} \leq 2 \Rightarrow x^2 + (y - 1)^2 \leq 4 \quad (27)$$

i.e. the circumference and the interior of the circle of radius 2, centered at the point $x = 0, y = 1$.

5.

(a) Let $z = x + iy$, $z^{-1} = a + ib$ (with a, b real). We want

$$zz^{-1} = (x + iy)(a + ib) = (xa - yb) + i(ya + xb) = 1 \quad (28)$$

giving the two simultaneous equations for a, b :

$$xa - yb = 1 \quad (29)$$

$$ya + xb = 0 \quad (30)$$

Multiply (29) by x and (30) by y and add:

$$a(x^2 + y^2) = x \quad (31)$$

Similarly

$$b(x^2 + y^2) = -y \quad (32)$$

By assumption $z \neq 0 \Rightarrow x^2 + y^2 \neq 0$. We then obtain the (unique) answer

$$a = \frac{x}{x^2 + y^2}, \quad b = \frac{-y}{x^2 + y^2} \Rightarrow z^{-1} = \frac{x - iy}{x^2 + y^2} \quad (33)$$

(b) $zz^{-1} = 1 \Rightarrow z^*(z^{-1})^* = 1$, and since the multiplicative inverse is unique, we can infer $(z^*)^{-1} = (z^{-1})^*$. Alternatively, complex-conjugating the answer for z^{-1} in part (a):

$$(z^{-1})^* = \frac{x + iy}{x^2 + y^2} = w^{-1} \quad (34)$$

where $w = x - iy = z^*$.

(c) Starting from the definition $\frac{z_1}{z_2} \equiv z_1 z_2^{-1}$:

$$\left(\frac{z_1}{z_2}\right)^* = (z_1 z_2^{-1})^* = z_1^* (z_2^{-1})^* = z_1^* (z_2^*)^{-1} = \frac{z_1^*}{z_2^*} \quad (35)$$

Another way is to first convince ourselves that division (starting from scratch) permits the operation

$$\frac{z_1}{z_2} = \frac{z_1 z}{z_2 z} \quad (36)$$

i.e. “multiplying top and bottom” by the same complex number. Consider $(zw)^{-1}$. By definition $(zw)(zw)^{-1} = 1$. But also $(zw)(z^{-1}w^{-1}) = (zz^{-1})(ww^{-1}) = 1$. By the uniqueness of the inverse:

$$(zw)^{-1} = z^{-1}w^{-1} \quad (37)$$

Then

$$\frac{z_1 z}{z_2 z} = (z_1 z)(z_2 z)^{-1} = z_1 z z_2^{-1} z^{-1} = z_1 z_2^{-1} = \frac{z_1}{z_2} \quad (38)$$

as required. We can now write

$$\frac{z_1}{z_2} = \frac{z_1 z_2^*}{z_2 z_2^*} = \frac{z_1 z_2^*}{|z_2|^2} \quad (39)$$

then complex-conjugate it:

$$\left(\frac{z_1}{z_2}\right)^* = \frac{z_1^* z_2}{|z_2|^2} \quad (40)$$

and, starting from the other end, compare with

$$\frac{z_1^*}{z_2^*} = \frac{z_1^* z_2}{z_2^* z_2} = \frac{z_1^* z_2}{|z_2|^2} \quad (41)$$

6.

(a) Recall that

$$\exp(-i\theta) = \cos(-\theta) + i \sin(-\theta) = \cos \theta - i \sin \theta \quad (42)$$

and thus $\exp(-i\theta)$ is the complex conjugate of $\exp(i\theta)$ (as expected, of course). The relations then follow from the general results $z + z^* = 2 \operatorname{Re} z$, $z - z^* = 2i \operatorname{Im} z$:

$$\exp(i\theta) + \exp(-i\theta) = (\cos \theta + i \sin \theta) + (\cos \theta - i \sin \theta) = 2 \cos \theta \quad (43)$$

$$\exp(i\theta) - \exp(-i\theta) = (\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta) = 2i \sin \theta \quad (44)$$

(b) Multiply top and bottom by $\exp(-i\theta/2)$:

$$\frac{\exp(i\theta) - 1}{\exp(i\theta) + 1} = \frac{\exp(i\theta/2) - \exp(-i\theta/2)}{\exp(i\theta/2) + \exp(-i\theta/2)} \quad (45)$$

$$= \frac{2i \sin(\theta/2)}{2 \cos(\theta/2)} = i \tan(\theta/2) \quad (46)$$

(c) Using the result above:

$$\frac{\exp(i\theta) + 1}{\exp(i\theta) - 1} = \frac{1}{i} \cot(\theta/2) \quad (47)$$

Add -1 to each side:

$$\frac{\exp(i\theta) + 1}{\exp(i\theta) - 1} - 1 = \frac{1}{i} \cot(\theta/2) - 1 \quad (48)$$

$$\Rightarrow \frac{2}{\exp(i\theta) - 1} = \frac{1}{i} \cot(\theta/2) - 1 \quad (49)$$

$$\Rightarrow \frac{1}{\exp(i\theta) - 1} = \frac{1}{2i} \cot(\theta/2) - \frac{1}{2} \quad (50)$$

7.

$$(1 + i)^{20} = 2^{10} \left(\frac{1 + i}{\sqrt{2}}\right)^{20} \quad (51)$$

$$= 2^{10} \exp(20i\pi/4) \quad (52)$$

$$= 2^{10} \exp(5\pi i) = -2^{10} \quad (53)$$